EC2200

Economics of Managerial Decision Making

Winter 2025 Examination · Paper 1

Duration 1.5 hours · Section A: 2 of 3 short questions (100 marks) · Section B: 1 of 2 long questions (100 marks)

Module	EC2200 — Economics of Managerial Decision Making
Session	Winter 2025
Structure	Section A: three short questions, answer any two (100 marks between them, 50 marks each). Section B: two long questions, answer one (100 marks). Both sections carry equal weight.
This scheme covers	All three Section A questions and both Section B questions in full.

Own-figure rule. If you carry an early error into a later part of a question but apply the right method consistently from that point on, you still pick up the marks for the later steps done correctly on your own figures.

A note on sourcing for this module. This is the first EC2200 marking scheme built for Study-Uni, and no lecture slides or module reading list were available to verify against. The content below follows the standard treatment of these topics in mainstream managerial economics texts (e.g. Baye & Prince, *Managerial Economics and Business Strategy*) and the named sources in the questions themselves (Adam Smith, Arrow 1963, Coase). If your lecturer's terminology, list of "five theories of profit", or specific diagram conventions differ from what's below, flag it and this can be adjusted — the underlying economics is standard, but course-specific framing can vary.

Section A · Question 1

Decision-Making Under Uncertainty, Share Valuation & Profit Maximisation · 15+15+20 marks

A1(a) **15 marks**

Name the possible outcomes resulting from a firm's decision which are necessary to consider that decision profitable.

A1(b) **15 marks**

A firm has expected profits per share of €10 in 2026, €15 in 2027, and €25 in 2028. The firm's investors require an 8% rate of return each year. Using the shareholder wealth-maximisation model, calculate the current (present) value of a share of stock for this firm. You must show your workings and round to two decimal places.

A1(c) **20 marks**

Let's say the profit (π) of a firm can be represented by the function below where Q represents the output level of the firm. Please find the value of Q and π when profit is maximized.

$$\pi = -50 + 160Q - 20Q^2$$

FULL MODEL ANSWER

(a) Outcomes necessary to assess whether a decision is profitable — 15 marks

Because almost every real managerial decision plays out under uncertainty, judging whether a decision is profitable requires more than just "the" revenue and "the" cost — it requires characterising the full range of ways the decision could turn out:

- **The possible states of the world / scenarios** the decision could result in (e.g. strong demand, weak demand, a competitor's response, a cost shock) — a decision is rarely certain to produce one single result.
- **The payoff (profit or loss) associated with each possible outcome** — the incremental revenue generated and the incremental cost incurred (including opportunity costs, not just explicit accounting costs) under each scenario.
- **The probability of each outcome occurring** — without a likelihood attached to each scenario, the possible payoffs alone cannot be combined into a single measure of whether the decision is worthwhile on average.
- **The firm's attitude to risk** — two decisions with the same expected (probability-weighted) payoff are not necessarily equally attractive if one has much more variable outcomes than the other; a risk-averse decision-maker may reject a positive-expected-value decision if the downside is severe enough.

> **Putting it together:** a decision is judged profitable by computing its **expected profit** — the probability-weighted sum of the payoffs across all possible outcomes — and comparing this (along with the risk involved) against the next-best alternative use of the firm's resources.

(b) Present value of a share — 15 marks

The shareholder wealth-maximisation model values a share as the present value of its expected future profits (or dividends per share), discounted at the investors' required rate of return:

$$V_0 = \Sigma \pi_t / (1+r)^t$$

Year	t	Profit per share (€)	Discount factor $1/(1.08)^t$	Present value (€)
2026	1	10.00	0.9259	9.26
2027	2	15.00	0.8573	12.86
2028	3	25.00	0.7938	19.85
Total present value (V_0)				**41.97**

> **Workings shown:** $V_0 = 10/(1.08)^1 + 15/(1.08)^2 + 25/(1.08)^3 = 9.26 + 12.86 + 19.85 =$ **€41.97**

(c) Profit-maximising output — 20 marks

Profit is maximised where marginal profit is zero (the first-order condition), confirmed by a negative second derivative (the second-order condition for a maximum rather than a minimum).

Step	Working
Profit function	$\pi = -50 + 160Q - 20Q^2$
First derivative (marginal profit)	$d\pi/dQ = 160 - 40Q$
Set equal to zero and solve	$\mathbf{160 - 40Q = 0 \Rightarrow Q^* = 4}$
Second derivative (check it's a maximum)	$d^2\pi/dQ^2 = -40 < 0 \Rightarrow$ confirmed maximum
Substitute Q*=4 into π	$\pi^* = -50 + 160(4) - 20(4)^2 = -50 + 640 - 320$
Profit-maximising output and profit	**$Q^* = 4$, $\pi^* =$ €270**

> **Don't skip the second-order condition:** setting the first derivative to zero only identifies a *stationary point* — it could be a maximum or a minimum. Confirming $d^2\pi/dQ^2 < 0$ is what establishes this is genuinely the profit-*maximising* output, and is routinely worth marks on its own.

Mark Allocation — Part (a) (15 marks)

Tier	Marks	What separates this tier
12–15	Identifies all key elements: possible outcomes/scenarios, the payoff of each, the probability of each, and the role of risk attitude, framed around expected profit.	
7–11	Identifies most elements (e.g. outcomes and probabilities) but misses risk attitude or the expected-value framing.	
0–6	Vague or generic discussion of 'costs and benefits' without engaging with uncertainty/outcomes specifically.	

Mark Allocation — Part (b) (15 marks)

Tier	Marks	What separates this tier
12–15	Correct formula, correct discount factors, correct individual present values, and the correct total (€41.97), with workings shown.	
7–11	Correct method with one arithmetic error carried through consistently (own-figure rule applies).	
0–6	Incorrect formula or method (e.g. failing to discount, or discounting at the wrong power of t).	

Mark Allocation — Part (c) (20 marks)

Tier	Marks	What separates this tier
16–20	Correct first derivative, correctly solved for $Q^*=4$, second-order condition checked and confirmed negative, and correct $\pi^*=270$.	
10–15	Correct Q^* and π^* but the second-order condition is not checked or is incorrect.	
0–9	Incorrect derivative or incorrect solving of the first-order condition.	

Question A1 Total: 50 Marks

Section A · Question 2

Demand, Supply, the Market Mechanism & Nudges · 15+15+20 marks

A2(a) 15 marks

Define and explain both the law of demand and the law of supply.

A2(b) 15 marks

Explain what the market mechanism is and its importance to economy.

A2(c) 20 marks

Detail and explain whether or not the use of nudges by firms has any implications for the law of demand. Use graphic illustrations if necessary.

FULL MODEL ANSWER

(a) The law of demand and the law of supply — 15 marks

The law of demand: holding all else constant (ceteris paribus), the quantity demanded of a good is inversely related to its price — as price rises, quantity demanded falls, and vice versa. This gives the standard downward-sloping demand curve, and reflects both a substitution effect (consumers switch to cheaper alternatives as price rises) and an income effect (a price rise reduces real purchasing power, further reducing quantity demanded for a normal good).

The law of supply: holding all else constant, the quantity supplied of a good is positively related to its price — as price rises, producers are willing to supply more, and vice versa. This gives the standard upward-sloping supply curve, reflecting rising marginal costs of production (producers need a higher price to justify producing additional, more costly units) and the greater profitability of supplying more at a higher price.

(b) The market mechanism — 15 marks

What it is: the market mechanism is the process by which the independent, decentralised decisions of many buyers and sellers, interacting through prices, coordinate to determine what is produced, how much, and at what price — without any central authority directing the outcome. Prices rise when quantity demanded exceeds quantity supplied (a shortage), which simultaneously encourages more supply and discourages some demand until the market clears; prices fall when there is a surplus, with the opposite effect.

Why it matters: the market mechanism allows resources to be allocated to their most highly valued uses using only decentralised price signals, without requiring any planner to know the enormous volume of information (individual preferences, production costs, available technology) held collectively across every buyer and seller in the economy. This is the foundation of the efficiency case for market-based allocation, and connects directly to Adam Smith's 'invisible hand' argument examined in Q. B1(a).

(c) Do nudges have implications for the law of demand? — 20 marks

Yes — nudges operate on demand, but not through the price mechanism the law of demand describes

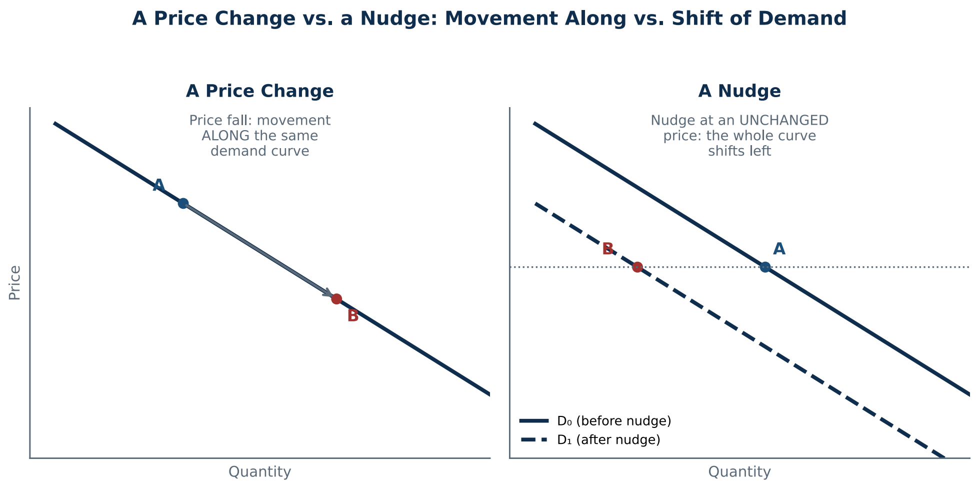

Fig. 1 — A price change moves along a fixed demand curve; a nudge shifts the whole curve at an unchanged price

The law of demand describes a movement ALONG a fixed curve: it isolates the effect of price, holding preferences, income, and everything else constant — a price fall causes a movement from point A to point B along the same demand curve (Fig. 1, left panel).

A nudge (Thaler & Sunstein) changes behaviour WITHOUT changing price: nudges work through choice architecture — default options, framing, salience, social proof — to influence what people choose, while leaving the price and the formal set of options unchanged (the definition of 'libertarian paternalism': choices remain freely available, just less likely to be taken up). A successful nudge that reduces demand for a product means that at the same price as before, consumers now buy less — this is not a movement along the demand curve, but a **leftward shift of the entire curve** (Fig. 1, right panel).

So the law of demand itself isn't violated or overturned: the inverse price-quantity relationship still holds at every point on the new (post-nudge) curve — a price cut would still raise quantity demanded relative to the new curve. What changes is the *position* of the demand curve, not the underlying law describing its slope.

Practical implication for firms: firms that want to influence demand have two distinct levers available: changing price (movement along the existing curve, governed by the law of demand and the price elasticity of demand) or changing the choice environment via nudges (shifting the curve itself, governed by behavioural factors outside the standard price-quantity relationship). Recognising which lever a given intervention pulls matters for predicting its effect correctly.

Exam tip: the strongest answers explicitly use the shift-vs-movement distinction with a labelled diagram, since this is precisely the concept the question is testing — a purely verbal answer that doesn't distinguish 'shift' from 'movement along' will struggle to reach the top band even if the general argument is sound.

Mark Allocation — Part (a) (15 marks)

Tier	Marks	What separates this tier
12–15	Correctly defines and explains both laws, including the ceteris paribus condition and the underlying economic reasoning (substitution/income effects for demand; marginal cost for supply).	
7–11	Correctly states both laws but the underlying reasoning is thin.	
0–6	One or both laws stated incorrectly or with major gaps.	

Mark Allocation — Part (b) (15 marks)

Tier	Marks	What separates this tier
12–15	Correctly explains the price adjustment process (shortage/surplus driving price changes) and articulates why this is important (decentralised information aggregation, efficient allocation).	
7–11	Correct basic description of the market mechanism but limited discussion of its importance.	
0–6	Vague or incorrect description.	

Mark Allocation — Part (c) (20 marks)

Tier	Marks	What separates this tier
16–20	Correctly distinguishes movement-along vs. shift-of the demand curve, correctly classifies a nudge as a curve-shifter rather than a price effect, uses a correct diagram, and draws out the practical implication for firms.	
10–15	Correctly identifies that nudges shift demand but the shift-vs-movement distinction is not fully or clearly explained, or no diagram is used.	
0–9	Minimal engagement with the shift-vs-movement distinction, or an incorrect claim that nudges violate the law of demand.	

Question A2 Total: 50 Marks

Section A · Question 3

Market Failure, Healthcare & the Irish Housing Market · 15+15+20 marks

A3(a) **15 marks**

Define market failure and list its four main causes.

A3(b) **15 marks**

Explain whether the argument set forth by Arrow (1963) would be in favour or opposed to healthcare being provided by the market.

A3(c) **20 marks**

Explain what could be done to help bring the Irish housing market into equilibrium.

FULL MODEL ANSWER

(a) Market failure and its four main causes — 15 marks

Definition: market failure occurs when the free market, left to its own devices, fails to allocate resources efficiently — the unregulated market outcome does not maximise total (social) welfare, so government intervention can, in principle, improve on the market outcome.

Cause	Why it breaks market efficiency
Externalities	A transaction imposes costs or benefits on third parties not party to it (e.g. pollution), so the price faced by buyers and sellers doesn't reflect the full social cost or benefit.
Public goods	Goods that are non-excludable and non-rivalrous (e.g. national defence, street lighting) cannot be efficiently provided by a market, since no individual firm can profitably charge for them and free-riding is unavoidable.
Market power (monopoly/imperfect competition)	A firm with market power restricts output and raises price above the competitive level to maximise its own profit, creating a deadweight loss relative to the competitive/efficient outcome.
Imperfect/asymmetric information	When buyers and sellers do not have the same information (e.g. about product quality or risk), markets can produce inefficient outcomes such as adverse selection or moral hazard.

(b) Arrow (1963) and market-provided healthcare — 15 marks

Arrow's argument is OPPOSED to healthcare being provided purely by the market

Arrow's core argument: in his 1963 paper 'Uncertainty and the Welfare Economics of Medical Care', Kenneth Arrow showed that the healthcare market has structural features that violate the conditions needed for a competitive market to be efficient, chiefly severe and pervasive **uncertainty** and **information asymmetry** between patients and providers.

- **Uncertainty of illness and treatment outcomes:** unlike most goods, the need for healthcare (and the effectiveness of any given treatment) is highly unpredictable, undermining the standard assumption of well-informed, repeat-purchase consumer decision-making.

- **Information asymmetry:** physicians possess vastly more medical knowledge than patients, meaning patients cannot easily judge the quality or necessity of the care they're being sold — a classic condition (Q. A3(a)) that a competitive market cannot handle well on its own.
- **Moral hazard:** because health insurance (a natural market response to the uncertainty above) shields patients from the full cost of care at the point of use, both patients and providers have weakened incentives to economise on treatment — a distortion inherent to insuring an uncertain, high-stakes good like health.
- **Adverse selection:** insurers cannot perfectly distinguish high-risk from low-risk individuals, which can undermine the viability of private insurance markets (the sickest are most eager to buy coverage, pushing up premiums and potentially driving out healthier, lower-risk buyers).

> **Conclusion Arrow draws:** because these market failures are not incidental but inherent to the nature of medical care, Arrow argues that non-market institutions — professional norms/licensing, non-profit provision, government regulation or provision, and social insurance — are a rational, efficiency-improving response to healthcare's unique characteristics, not merely a redistributive or ideological choice. This directly opposes the view that healthcare should be left to an unregulated market.

(c) Bringing the Irish housing market into equilibrium — 20 marks

Applying the supply-and-demand framework from Q. A2: a persistent shortage (excess demand at prevailing prices, reflected in rapidly rising prices and rents) means either supply needs to increase, demand needs to moderate, or both, for the market to move toward equilibrium.

> **Supply-side measures (the most direct route to equilibrium):** streamlining planning permission and reducing approval timelines; increasing the release of zoned, serviced land; addressing construction sector capacity constraints (labour supply, cost of building materials, productivity in the construction sector); and incentivising institutional and private investment in new housing supply (build-to-rent, cost-rental models) to add units directly to the housing stock.

> **Demand-side considerations (a note of caution):** many existing Irish housing policies work through the demand side — schemes such as Help-to-Buy or the First Home Scheme increase buyers' purchasing power. Without a matching increase in supply, demand-side subsidies risk simply bidding up prices further (shifting the demand curve right against an unresponsive supply curve), rather than closing the gap between quantity demanded and quantity supplied.

> **Why the supply side is the binding constraint:** if supply is highly inelastic in the short run (as is typical for housing, given long construction lead times and planning processes), then most of any demand-side stimulus shows up as higher prices rather than higher quantity built — consistent with the standard supply-and-demand diagram from Q. A2(a): a demand shift against a steep (inelastic) supply curve produces a large price change and only a small quantity change.

> **A balanced policy mix:** the most effective route to equilibrium combines supply-side reform (the dominant lever, given Ireland's persistent shortfall in completions relative to underlying housing need) with demand-side measures carefully targeted so as not to outpace the supply response, plus addressing specific frictions (e.g. rental market regulation, vacancy, and the interaction between housing and broader macroeconomic/immigration-driven demand growth).

Link back to Q. A2(b): a well-functioning market mechanism relies on price signals eliciting a *supply* response as well as rationing demand — the core diagnosis of the Irish housing market is that supply has been unable to respond adequately to persistently high prices (due to planning, land, and construction capacity constraints), which is exactly why supply-side reform, not price controls or demand stimulus alone, is generally identified as the priority.

Mark Allocation — Part (a) (15 marks)

Tier	Marks	What separates this tier
12–15	Correct definition and all four causes (externalities, public goods, market power, information asymmetry) correctly named and briefly explained.	
7–11	Correct definition with two or three causes correctly identified.	
0–6	Incorrect or incomplete definition, fewer than two causes correctly identified.	

Mark Allocation — Part (b) (15 marks)

Tier	Marks	What separates this tier
12–15	Correctly identifies Arrow's position as opposed to pure market provision, with at least two of the specific mechanisms (uncertainty, information asymmetry, moral hazard, adverse selection) correctly explained.	
7–11	Correct overall position (opposed) with one mechanism explained in reasonable depth.	
0–6	Incorrect position, or no engagement with Arrow's specific argument.	

Mark Allocation — Part (c) (20 marks)

Tier	Marks	What separates this tier
16–20	Identifies supply-side measures as the primary lever, explains why (inelastic short-run supply), discusses the risk that demand-side measures alone worsen prices, and connects the analysis back to the supply-and-demand framework.	
10–15	Identifies reasonable supply- and/or demand-side measures without a clear supply-vs-demand analytical framing.	
0–9	Vague or purely descriptive policy list without economic reasoning.	

Question A3 Total: 50 Marks

Section B: Long Questions

Answer 1 of 2 · this scheme covers both in full

Section B questions are broader and typically integrate several ideas together.

Section B · Question 1(a)

Adam Smith and the Invisible Hand · 30 marks

B1(a) **30 marks**

As per the work of Adam Smith, explain how the pursuit of profit by firms can produce beneficial outcomes for society.

FULL MODEL ANSWER

The 'invisible hand' argument

In *The Wealth of Nations* (1776), Adam Smith argued that individuals and firms pursuing their own self-interest — profit, in the case of a firm — are, without intending to, led "as if by an invisible hand" to promote outcomes that benefit society as a whole, even though that was no part of their intention.

The mechanism, step by step

> **Profit as a signal:** profit tells a firm that consumers value what it produces more than the resources used to produce it cost — a profitable activity is, in a well-functioning market, one that is creating more value than it consumes. Firms chasing profit are thereby drawn toward activities that genuinely create value for others, not away from them.

> **Competition disciplines self-interest:** a firm's pursuit of profit does not go unchecked — if a firm charges too much or produces poor-quality goods, competitors have a profit incentive to undercut it on price or out-compete it on quality, disciplining the pursuit of self-interest into outcomes that benefit consumers (lower prices, better products) rather than pure exploitation.

> **Specialisation and the division of labour:** the pursuit of profit drives firms and individuals to specialise in what they can produce most efficiently (comparative advantage) and trade the surplus — Smith's famous pin factory example shows how this dramatically raises total output compared to each worker/firm trying to be self-sufficient, benefiting society through greater overall production from the same resources.

> **Resources flow to their most valued uses:** the pursuit of profit continuously redirects capital and labour away from activities that are no longer profitable (i.e. no longer creating enough value relative to cost) and toward those that are — the market mechanism (Q. A2(b)) that channels self-interested decisions into an efficient allocation of society's scarce resources, without any central planner directing the process.

The key insight: no benevolence required

Smith's argument is deliberately not a claim that firms are, or need to be, altruistic. His point is that a well-functioning competitive market <u>channels</u> self-interested profit-seeking into socially beneficial outcomes as a by-product — "It is not from the benevolence of the butcher, the brewer, or the baker that we expect our dinner, but from their regard to their own interest." The beneficial social outcome emerges from the structure of market competition itself, not from firms' intentions.

An important qualification, worth noting for a complete answer: Smith's argument depends on the market actually being competitive and reasonably well-functioning — where market failures are present (Q. A3(a): externalities, market power, public goods, information asymmetry), the pursuit of profit can instead produce socially harmful outcomes (e.g. the pollution case in Q. B2(c)), since the invisible-hand mechanism relies on prices correctly reflecting full social costs and benefits.

Mark Allocation (30 marks)

Tier	Marks	What separates this tier
24–30	Explains the invisible-hand mechanism with at least three distinct supporting elements (profit as a value signal, competition as a discipline, specialisation/division of labour, efficient resource allocation), and correctly notes that no benevolence is required — the outcome emerges from market structure.	
15–23	Explains the general invisible-hand idea with one or two supporting elements in reasonable depth.	
0–14	Vague reference to 'the invisible hand' without explaining the underlying mechanism.	

Section B · Question 1(b)

The Five Theories of Profit · 30 marks

B1(b) **30 marks**

List and briefly explain the five theories of profit.

FULL MODEL ANSWER

Economists have long asked why profit exists at all in a fully competitive market (where, in the textbook long-run competitive equilibrium, economic profit is driven to zero). The five classic theories offer different answers — each identifying a different source of persistent, above-normal profit.

Theory	Explanation
1. Frictional (compensatory) theory	Markets are not always in equilibrium — unexpected changes in demand or supply create temporary imbalances, and firms that happen to be well-positioned when the shock hits earn above-normal profit until the market adjusts back toward equilibrium. Profit here is a short-run, self-correcting phenomenon.
2. Monopoly theory	Firms with market power — through barriers to entry, patents, exclusive licences, or control of a key input — can restrict output and sustain a price above the competitive level indefinitely, earning persistent above-normal profit that isn't competed away.
3. Innovation theory (Schumpeter)	Profit is the reward earned by a firm that successfully introduces a new product, process, or way of doing business ('creative destruction') ahead of its competitors — a temporary reward for innovation that persists only until competitors imitate or the innovation is superseded.
4. Risk-bearing (compensatory) theory (Knight)	Profit compensates entrepreneurs and firms for bearing genuine, uninsurable *uncertainty* (as distinct from measurable, insurable risk) — those willing to commit capital to uncertain ventures are compensated, on average, with a return above what a safe investment would offer.
5. Managerial efficiency theory	Some firms simply manage their resources more effectively than their competitors — better cost control, superior organisation, more effective decision-making — and earn above-normal profit as a direct result of this superior efficiency, even operating in an otherwise competitive market.

Terminology note: different textbooks group and label these slightly differently (e.g. some combine frictional and risk-bearing under a broader 'disequilibrium/compensatory' heading, or list a separate 'rent' theory of profit for scarce factors). If your course materials use different names for the same five ideas, the underlying economic content above should still map across — flag any specific terminology mismatch and this can be adjusted.

Mark Allocation (30 marks, 6 per theory)

Tier	Marks	What separates this tier
5–6 per theory	Theory correctly named and the underlying economic mechanism (why profit arises and whether it's temporary or persistent) correctly explained.	
2–4 per theory	Theory correctly named with a partial or thin explanation.	
0–1 per theory	Theory not named or incorrectly explained.	

Section B · Question 1(c)

The Decision-Making Process & MR=MC · 40 marks

B1(c) **40 marks**

With the aid of a diagram and reference to the decision-making model of the firm to increase profit, explain the process by which a firm can achieve its objectives.

FULL MODEL ANSWER

The managerial decision-making process

Achieving the firm's objective (profit maximisation) is not a single calculation but a structured process:

- **Step 1 — Define the objective and the decision to be made.** Typically profit maximisation (or, in the shareholder wealth-maximisation model of Q. A1(b), maximising the present value of the firm), applied to a specific decision (how much to produce, what price to charge, whether to enter a market).
- **Step 2 — Identify the constraints the firm faces.** Technology/production capacity, input costs, competitors' behaviour, and market demand for the firm's product all constrain which outputs are actually achievable.
- **Step 3 — Identify the feasible alternatives.** The range of output levels, prices, or strategies the firm could realistically choose given its constraints.
- **Step 4 — Evaluate the alternatives at the margin.** For each additional unit of output, compare the marginal revenue (MR) it would generate against the marginal cost (MC) of producing it — this marginal, incremental comparison, rather than an average or total comparison, is the core analytical tool of managerial decision-making.
- **Step 5 — Choose the optimal alternative and implement it.** Select the output/price combination that maximises the objective, implement the decision, and monitor outcomes to inform future decisions (feeding back into Step 1 for the next decision).

The core analytical tool: MR = MC

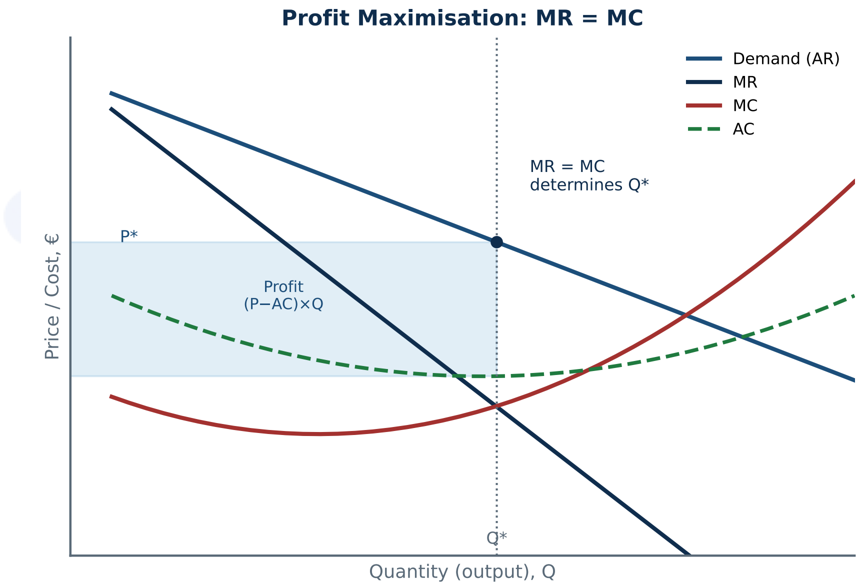

Fig. 1 — Profit is maximised where MR = MC; the shaded area shows the resulting profit (P−AC)×Q at that output

> **Why MR = MC identifies the profit-maximising output:** as long as an additional unit adds more to revenue than it adds to cost (MR > MC), producing it raises total profit, so the firm should keep expanding output. Once an additional unit would cost more than it earns (MR < MC), producing it would reduce profit, so the firm should stop short of that unit. The profit-maximising output is therefore exactly where MR = MC (Fig. 1) — this is the same marginal-analysis logic used to solve Q. A1(c) algebraically (setting dπ/dQ = 0 is mathematically identical to setting MR = MC, since dπ/dQ = MR − MC).

> **Price and profit follow from Q*:** once the profit-maximising quantity (Q*) is identified from the MR=MC intersection, the price the firm can charge for that quantity is read off the demand curve (P* in Fig. 1), and total profit is the resulting gap between price and average cost (AC), multiplied by quantity — the shaded rectangle in Fig. 1.

Bringing the process together

The five-step decision process (define objective → identify constraints → identify alternatives → evaluate at the margin → choose and implement) is the general managerial framework; the MR=MC condition is the specific marginal-analysis tool that operationalises Step 4 for the standard output/pricing decision. Together, they describe how a firm moves from a general profit-maximisation objective to a concrete, optimal output and pricing decision.

> **Exam tip:** a strong answer for this part explicitly connects the general decision-making *process* (the five steps) to the specific *tool* (MR=MC) that implements it — simply drawing the MR=MC diagram without discussing the broader decision process (or vice versa) leaves out half of what the question asks for.

Mark Allocation (40 marks)

Tier	Marks	What separates this tier
30–40	Explains the full decision-making process (objective, constraints, alternatives, marginal evaluation, implementation) AND correctly presents and explains the MR=MC diagram, explicitly linking the two.	
20–29	Covers either the decision-making process OR the MR=MC diagram in good depth, but not both, or the link between them is not made explicit.	
10–19	Partial or generic discussion of 'firms want to maximise profit' without a structured process or a correct diagram.	
0–9	Minimal or largely incorrect treatment.	

Question B1 Total: 100 Marks (30 + 30 + 40)

Section B · Question 2(a)

The Fundamental Assumptions of Utility Theory · 30 marks

B2(a) **30 marks**

List and explain the fundamental assumptions underlying utility theory.

FULL MODEL ANSWER

Assumption	Explanation
1. Completeness	A consumer can always rank any two bundles of goods — for any A and B, the consumer either prefers A to B, prefers B to A, or is indifferent between them. Preferences are never 'undefined' when comparing two options.
2. Transitivity	If a consumer prefers A to B, and prefers B to C, then they must prefer A to C. This ensures preferences are internally consistent and can be represented by a well-behaved ranking, ruling out preference cycles (A > B > C > A).
3. Non-satiation (more is better)	More of a good is always at least as good as, and generally strictly preferred to, less of it — consumers are never fully 'satisfied' to the point where an additional unit of a good makes them worse off. This underpins upward-sloping indifference curve maps and ensures utility functions are increasing in each good.
4. Diminishing marginal utility / convexity of preferences	As a consumer acquires more of a given good, the additional (marginal) utility from each extra unit diminishes — and consumers prefer balanced bundles of goods to extreme ones (a mix of two goods is preferred to all of one and none of the other, for bundles the consumer is indifferent between). This gives standard indifference curves their convex, bowed-toward-the-origin shape.

> **Why these assumptions matter:** together, completeness and transitivity guarantee that a consumer's preferences can be represented by a well-defined utility function at all; non-satiation and diminishing marginal utility/convexity are what give that utility function its standard, familiar shape (increasing, with diminishing returns) and generate the conventional downward-sloping demand curve (Q. A2(a)) once combined with a budget constraint.

If your course also covers expected utility theory (choice under uncertainty, directly relevant to Q. A1(a)'s discussion of risky decisions), a further assumption — the **independence axiom** — is often added: preferences between two risky prospects should not be affected by mixing both with an identical third prospect. This is a more advanced addition specific to decision-making under risk, distinct from the four core assumptions above which apply to standard consumer choice under certainty.

Mark Allocation (30 marks)

Tier	Marks	What separates this tier
24–30	Correctly lists and explains at least four core assumptions (completeness, transitivity, non-satiation, diminishing marginal utility/convexity), with correct economic reasoning for each.	
15–23	Lists two to three assumptions with reasonable explanation.	
0–14	Lists fewer than two assumptions correctly, or explanations are largely incorrect.	

Section B · Question 2(b)

Cigarette Tax: Pigouvian Tax or Nudge? · 30 marks

B2(b) **30 marks**

The government wishes to place a 50% tax on the sale of cigarettes to reduce the negative consequences addictive smoking has for health. Clearly state whether this policy initiative is a nudge or a Pigouvian tax and explain the difference between the two.

FULL MODEL ANSWER

This is a Pigouvian tax, not a nudge

Why this is a Pigouvian tax

Definition: a Pigouvian tax is a tax set (ideally) equal to the marginal external cost a good imposes on third parties/society, designed to make the private cost faced by the consumer reflect the full social cost, correcting the over-consumption that arises from a negative externality.

Applying it here: smoking imposes costs beyond the smoker themselves — on the public health system, through second-hand smoke on others, and arguably through internalities the smoker doesn't fully weigh due to addiction. A 50% tax directly and substantially raises the price smokers pay, working through the standard law of demand (Q. A2(a)): a higher price reduces quantity demanded, moving consumption along the demand curve toward the socially optimal (lower) quantity.

Why this is NOT a nudge

What a nudge is: from Q. A2(c), a nudge (Thaler & Sunstein's 'libertarian paternalism') changes behaviour through choice architecture — defaults, framing, salience, social norms — while leaving the economic incentives (prices) and the formal choice set unchanged. A nudge shifts the demand curve; it does not move price.

This policy is the opposite of that: a 50% tax is a direct, substantial, mandatory change to the price of cigarettes — exactly the kind of price-based intervention that a nudge is specifically defined in contrast to. There's no preservation of the underlying incentive structure here; the whole point of the policy is to change it.

The key distinction, summarised

	Pigouvian tax	Nudge
Mechanism	Changes relative prices to internalise an externality	Changes choice architecture without changing prices or options
Effect on demand curve	Movement ALONG the demand curve (price change)	SHIFT of the demand curve (price unchanged)
Coercion / restriction of choice	Consumer can still buy the good, just at a higher price (financial cost imposed)	Consumer retains the exact same options at the exact same prices (behavioural, not financial, influence)
This policy (50% cigarette tax)	**Yes — this is a Pigouvian tax**	**No**

Exam tip: the question specifically asks you to state clearly which one this is before explaining the difference — make sure the verdict is unambiguous and stated early, since a well-argued but ambiguous answer (or one that hedges between the two) will lose marks even if the surrounding explanation of both concepts is otherwise strong.

Mark Allocation (30 marks)

Tier	Marks	What separates this tier
24–30	Clear, unambiguous verdict (Pigouvian tax) stated early, correct definition of both a Pigouvian tax and a nudge, and a clear explanation of the underlying distinction (price change/movement along the curve vs. choice-architecture change/shift of the curve).	
15–23	Correct verdict with a reasonable explanation of one of the two concepts, but the distinction between them is not fully drawn out.	
0–14	Incorrect verdict, or no clear distinction drawn between the two concepts.	

Section B · Question 2(c)

Marine Pollution & the Limits of Coase's Theorem · 40 marks

B2(c) **40 marks**

A pharmaceutical firm chooses to dispose of their waste in a public marine ecosystem. Locals and other firms who derive utility from the marine ecosystem object to this but cannot reach a satisfactory resolution with the polluting firm. With reference to Coase’s theorem, explain why a resolution in this case may be difficult to reach without government intervention.

FULL MODEL ANSWER

Coase's theorem, briefly

Coase's theorem states that, in the presence of an externality (here, pollution — a negative externality, Q. A3(a)), if **property rights are clearly defined**, **transaction costs are low**, and all parties have **full information**, private bargaining between the affected parties will reach an efficient outcome regardless of who initially holds the property right — without needing government intervention (e.g. a Pigouvian tax, Q. B2(b)) to correct the externality.

Why this case violates each condition

1. Property rights over the marine ecosystem are not well-defined. a marine ecosystem is a classic common/public resource (Q. A3(a)) — no single party (the locals, the other firms, or the pharmaceutical firm) holds a clear, exclusive, and enforceable property right over it. Coase's theorem specifically requires a well-defined right to bargain over; without one, it is unclear what exactly is being negotiated (a right to pollute? a right to a clean ecosystem? whose right is it to sell or buy?).

2. Transaction costs are high due to the number of affected parties. the theorem works most cleanly in a two-party bargain. Here, there are many affected parties (“locals and other firms”) with potentially differing valuations of the ecosystem, all of whom would need to coordinate to negotiate collectively with the polluting firm. Organising this many-party negotiation — agreeing a common position, avoiding free-riding (some locals might let others bear the cost of negotiating while still enjoying any resulting benefit), and reaching unanimous agreement — is costly and difficult in practice, exactly the kind of transaction cost Coase's theorem assumes away.

3. Information problems. accurately valuing the damage to the marine ecosystem (and how that damage is distributed across many different locals and firms with different uses of it) requires information that may not be readily available or agreed upon, making it hard even to know what an efficient bargain would look like, let alone reach one.

The conclusion

Because this case violates all three of Coase's key conditions (undefined property rights, high transaction costs from the multi-party setting, and information problems), private bargaining between the locals, the other firms, and the polluting firm is unlikely to reach the efficient outcome on its own — consistent with the scenario description, where a resolution has not been reached. This is precisely the situation in which economic theory points to a role for **government intervention**: for example, clearly assigning and enforcing a property right over the ecosystem (e.g. designating it as protected and prohibiting dumping), imposing a Pigouvian tax on the pollution equal to its external cost, or issuing and enforcing tradable pollution permits — any of which can substitute for the private bargain that high transaction costs and undefined rights are preventing from happening here.

Exam tip: the question specifically asks you to explain why resolution is *difficult* using Coase's theorem — the strongest answers work through the theorem's actual conditions (property rights, transaction costs, information) one by one and show how *this specific scenario* violates each, rather than just asserting 'the market fails here' without connecting it back to the theorem's structure.

Mark Allocation (40 marks)

Tier	Marks	What separates this tier
30–40	Correctly states Coase's theorem and its three conditions, and systematically explains how the scenario violates each (undefined property rights over a common resource, high transaction costs from many affected parties, information problems), concluding with the role for government intervention.	
20–29	Correctly states Coase's theorem and identifies one or two of the violated conditions in reasonable depth.	
10–19	General reference to Coase's theorem or 'market failure' without working through the specific conditions.	
0–9	Minimal or incorrect treatment.	

Question B2 Total: 100 Marks (30 + 30 + 40)